

MIAE 221: Materials Science for Engineers

Chapters 18, 19 & 21: Electrical, Thermal and Optical Properties (Expanded & Condensed Guide)

Concordia University · Gina Cody School of Engineering

Textbook:

Materials Science and Engineering: An Introduction

(10th Ed., Callister & Rethwisch)

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(Syllabus Week 11 · Final Exam Scope)

1. 🎯 30-Second Core Intuition & Plain-English Concept

Electrical conduction depends on whether electrons can easily jump into empty, available energy states to move freely through the solid.

Energy Band Theory:

- **Metals:** Conduction and valence bands overlap (or band is half-filled). Billions of electrons are free to move instantly $\implies$ high conductivity.
- **Insulators:** A massive forbidden energy band gap ($E_g > 5\text{ eV}$) separates filled and empty states. Electrons cannot jump $\implies$ zero conductivity.
- **Semiconductors:** A narrow band gap ($E_g < 2\text{ eV}$). Thermal energy can kick electrons across the gap $\implies$ conductivity increases exponentially with temperature!

2. ⚙️ High-Yield Mathematical Engine & Semiconductor Physics

1. Ohm's Law and Electrical Conductivity

$$V = IR, \quad \rho_{\text{el}} = \frac{RA}{l}, \quad \sigma_{\text{el}} = \frac{1}{\rho_{\text{el}}} = n|e|\mu_e + p|e|\mu_h$$

- n, p : Free electron and hole concentrations (extm^{-3}).
- μ_e, μ_h : Electron and hole mobilities ($\text{extm}^2/\text{V} \cdot \text{s}$).
- e : Elementary charge ($1.602 \times 10^{-19} \text{ C}$).

2. Intrinsic vs. Extrinsic Semiconductors

- **Intrinsic (Pure Si, Ge):** $n = p = n_i$. Conductivity rises exponentially with temperature:

$$\sigma \propto \exp\left(-\frac{E_g}{2k_B T}\right)$$

- **Extrinsic (Doped):**

- **n -Type:** Doped with Group V elements (P, As, Sb). Donates free electrons ($n \gg p$).
- **p -Type:** Doped with Group III elements (B, Al, Ga). Creates positive holes ($p \gg n$).

3. Textbook Reference Diagram & Visual Pedagogical Analysis

Figure 18.4 The various possible electron band structures in solids at 0 K. (a) The electron band structure found in metals such as copper, in which there are available electron states above and adjacent to filled states, in the same band. (b) The electron band structure of metals such as magnesium, in which there is an overlap of filled and empty outer bands. (c) The electron band structure characteristic of insulators; the filled valence band is separated from the empty conduction band by a relatively large band gap (>2 eV). (d) The electron band structure found in the semiconductors, which is the same as for insulators except that the band gap is relatively narrow (<2 eV).

The number of states within each band is equal to the total of all states contributed by the N atoms. For example, an s band consists of N states and a p band of $3N$ states. With regard to occupancy, each energy state may accommodate two electrons that must have oppositely directed spins. Furthermore, bands contain the electrons that resided in the corresponding levels of the isolated atoms; for example, a $4s$ energy band in the solid contains those isolated atoms' $4s$ electrons. Of course, there are empty bands and, possibly, bands that are only partially filled.

The electrical properties of a solid material are a consequence of its electron band structure—that is, the arrangement of the outermost electron bands and the way in which they are filled with electrons.

Four different types of band structures are possible at 0 K. In the first (Figure 18.4a), one outermost band is only partially filled with electrons. The energy corresponding to the highest filled state at 0 K is called the **Fermi energy** E_F , as indicated. This energy band structure is typified by some metals, in particular those that have a single s valence electron (e.g., copper). Each copper atom has one $4s$ electron; however, for a solid composed of N atoms, the $4s$ band is capable of accommodating $2N$ electrons. Thus, only half of the available electron positions within this $4s$ band are filled.

For the second band structure, also found in metals (Figure 18.4b), there is an overlap of an empty band and a filled band. Magnesium has this band structure. Each

Fermi energy

Figure 18.4: Electron band structures: (a, b) Metals (overlapping / partially filled), (c) Insulators (wide gap), and (d) Semiconductors (narrow gap).

In-Depth Visual Breakdown:

- **Figure 18.4:** Demonstrates why metals conduct and insulators do not:
 - In metals (a, b), the Fermi level E_F lies inside a band or between overlapping bands. Adjacent unoccupied states exist immediately above E_F ; negligible electrical field excites electrons into motion.
 - In insulators (c), the valence band is completely full and separated from the empty conduction band by a wide forbidden gap ($E_g > 5 \text{ eV}$). Room temperature thermal energy ($k_B T \approx 0.026 \text{ eV}$) is completely incapable of exciting electrons across this gap.
 - In semiconductors (d), $E_g \approx 1.1 \text{ eV}$ (Silicon). Thermal vibrations easily kick a steady stream of electrons into the conduction band, leaving equal numbers of conducting holes behind.

4. 🏛️ Teacher Notes Cross-Reference & Concordia Exam Traps

- **Concordia Exam Focus:**
 - Explaining the opposing temperature effects:
 - In **metals**, electrical conductivity **decreases** as temperature rises (thermal lattice vibrations scatter electrons: resistivity $\rho = \rho_0 + \alpha T$).
 - In **intrinsic semiconductors**, electrical conductivity **increases exponentially** with temperature (thermal energy excites vastly more charge carriers across E_g).

📖 Chapter 19: Thermal Properties

(Syllabus Week 11 · Final Exam Scope)

1. 🎯 30-Second Core Intuition & Plain-English Concept

Heat is simply atomic vibrations. In solids, heat is stored in quantized vibrational waves called **phonons** and transported through the material by both phonons and free electrons.

Why Materials Expand When Heated: Thermal expansion is a direct consequence of the **asymmetry (anharmonicity)** of the interatomic potential energy well! Because it takes less energy to push atoms apart than to squeeze them together, heating an atom makes it vibrate outward, increasing the average interatomic spacing.

2. ⚙️ High-Yield Mathematical Engine & Thermal Properties

1. Heat Capacity (C_v)

$$C_v = \left(\frac{\partial E}{\partial T} \right)_v$$

- At high temperatures ($T > \theta_D$, Debye Temperature), heat capacity reaches the universal **Dulong-Petit Limit**:

$$C_v \approx 3R \approx 25 \text{ J/mol} \cdot \text{K}$$

2. Linear Thermal Expansion

$$\frac{\Delta l}{l_0} = \alpha_l \Delta T$$

- α_l : Linear coefficient of thermal expansion (extK^{-1} or $^{\circ}\text{extC}^{-1}$).
- Ceramics have lower α_l than metals; polymers have the highest α_l (weak van der Waals bonds).

3. Thermal Conductivity (k) and Thermal Shock Resistance (TSR)

$$q = -k \frac{dT}{dx} \quad (\text{Fourier's Law})$$

- **Thermal Shock Resistance (TSR):** The ability of a ceramic to withstand rapid cooling without fracture:

$$TSR \cong \frac{\sigma_f \cdot k}{E \cdot \alpha_l}$$

(To resist cracking during quenching, a ceramic must have high strength σ_f , high thermal conductivity k , low stiffness E , and low thermal expansion α_l)

3. 📖 Textbook Reference Diagram & Visual Pedagogical Analysis

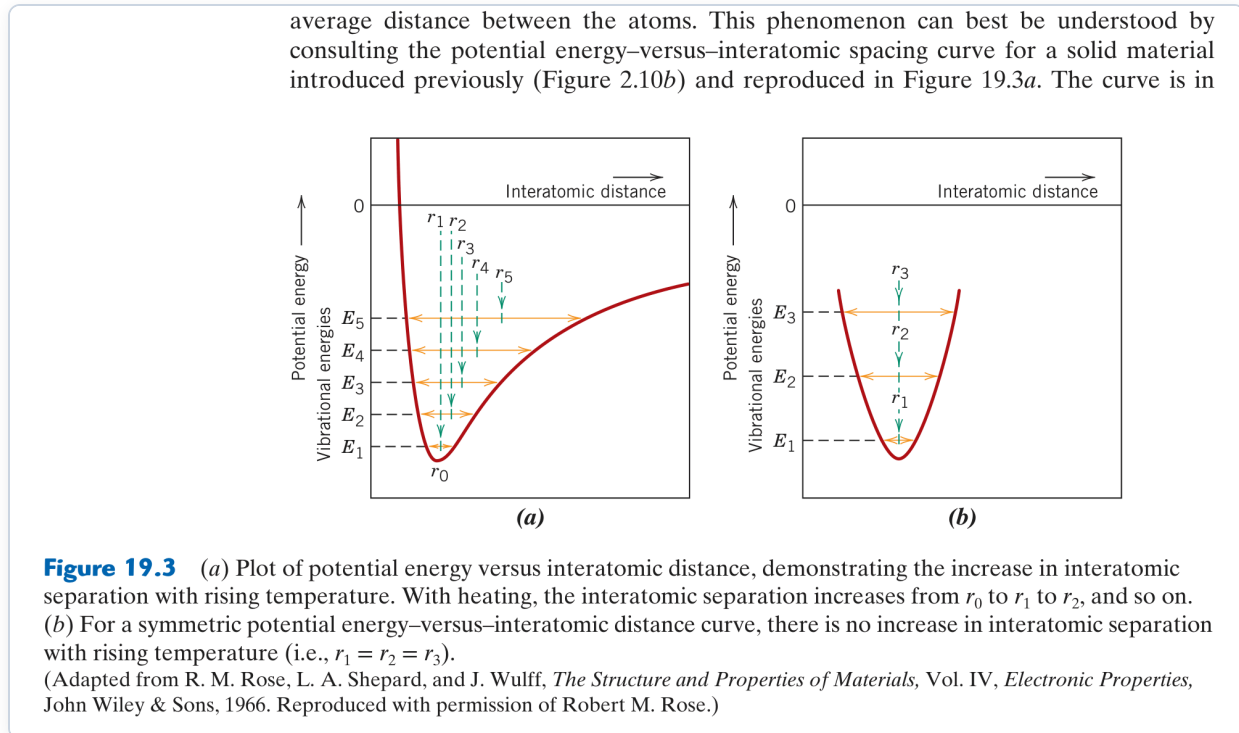


Figure 19.3: Potential energy vs. interatomic distance: (a) Asymmetric well demonstrating increase in average interatomic spacing r_T with rising temperature, vs. (b) A hypothetical symmetric well with zero thermal expansion.

In-Depth Visual Breakdown:

- **The Asymmetric Well (Figure 19.3a):** At $0\text{ext}K$, atoms sit at minimum r_0 . As temperature rises to T_1, T_2, T_3 , the atom vibrates between the left and right walls of the well. Because the repulsive wall (left) is much steeper than the attractive wall (right), the midpoint of vibration shifts progressively outward ($r_3 > r_2 > r_1 > r_0$). This macroscopic shift is thermal expansion.
- **The Hypothetical Symmetric Well (Figure 19.3b):** If interatomic forces were perfectly symmetric (parabolic well), heating would expand vibrations equally in both directions, the average separation would remain constant ($r_T = r_0$), and the material would have **zero thermal expansion** ($\alpha_l = 0$).

📖 Chapter 21: Optical Properties

1. 🎯 30-Second Core Intuition & Plain-English Concept

Light is electromagnetic radiation. When a light wave strikes a solid, three things can happen: it can bounce off (**Reflection**), get absorbed by electrons (**Absorption**), or pass through unscathed (**Transmission**):

$$R + A + T = 1$$

Why Glass is Transparent and Metals are Opaque:

- **Metals:** Have empty electronic energy states everywhere. Photons of *all* visible wavelengths are instantly absorbed by electrons and re-emitted $\implies$ metals are totally opaque and highly reflective.
- **Insulators / Glasses:** Have an energy band gap ($E_g > 3.1\text{eV}$) that is larger than visible light photon energies ($1.8 - 3.1\text{eV}$). Photons do not possess enough energy to excite electrons across the band gap $\implies$ light cannot be absorbed and passes straight through (transparent!).

2. ⚙️ High-Yield Mathematical Engine & Optical Physics

1. Photon Energy and Wavelength Relation

$$E = h\nu = \frac{hc}{\lambda}$$

- h : Planck's constant ($4.136 \times 10^{-15} \text{ eV} \cdot \text{s} = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$).
- c : Speed of light ($3.0 \times 10^8 \text{ m/s}$).
- **Concordia Shortcut:**

$$E(\text{eV}) = \frac{1.24}{\lambda(\mu\text{m})}$$

2. Fundamental Absorption Condition

A photon can be absorbed by an electron transition across a band gap if and only if:

$$h\nu \geq E_g \iff \lambda \leq \frac{hc}{E_g}$$

- If $E_g > 3.1 \text{ eV}$ ($\lambda < 0.4 \mu\text{m}$, ultraviolet), visible light ($0.4 - 0.7 \mu\text{m}$) cannot be absorbed, making the material **optically transparent** (e.g., Diamond $E_g = 5.5 \text{ eV}$).

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

Optical Properties of Metals

Consider the electron energy band schemes for metals as illustrated in Figures 18.4a and 18.4b; in both cases, a high-energy band is only partially filled with electrons. Metals are opaque because the incident radiation having frequencies within the visible range excites electrons into unoccupied energy states above the Fermi energy, as demonstrated in Figure 21.4a; as a consequence, the incident radiation is absorbed, in accordance with Equation 21.6. Total absorption is within a very thin outer layer, usually less than $0.1\ \mu\text{m}$; thus only metallic films thinner than $0.1\ \mu\text{m}$ are capable of transmitting visible light.

All frequencies of visible light are absorbed by metals because of the continuously available empty electron states, which permit electron transitions as in Figure 21.4a. In fact, metals are opaque to all electromagnetic radiation on the low end of the frequency

Figure 21.4 (a) Schematic representation of the mechanism of photon absorption for metallic materials in which an electron is excited into a higher-energy unoccupied state. The change in energy of the electron ΔE is equal to the energy of the photon. (b) Reemission of a photon of light by the direct transition of an electron from a high to a low energy state.

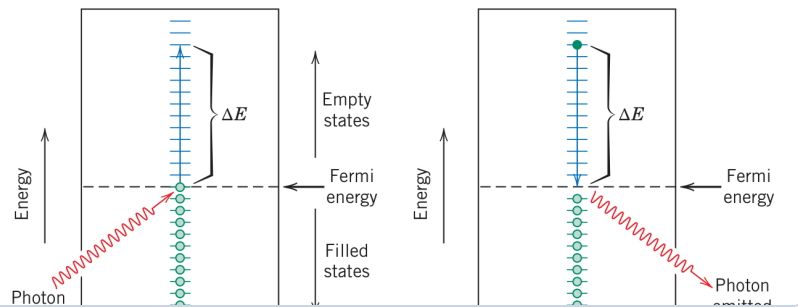


Figure 21.4: (a) Electron excitation across a band gap via photon absorption ($h\nu > E_g$), and (b) Subsequent decay and light re-emission.

In-Depth Visual Breakdown:

- **Photon Absorption (Figure 21.4a):** An incoming photon with energy $h\nu \geq E_g$ is annihilated, promoting an electron from the valence band into the conduction band and creating a hole in the valence band.
- **Electron Relaxation (Figure 21.4b):** When the excited electron falls back to the valence band, it releases energy by emitting a photon (luminescence/fluorescence) or generating heat (phonons).